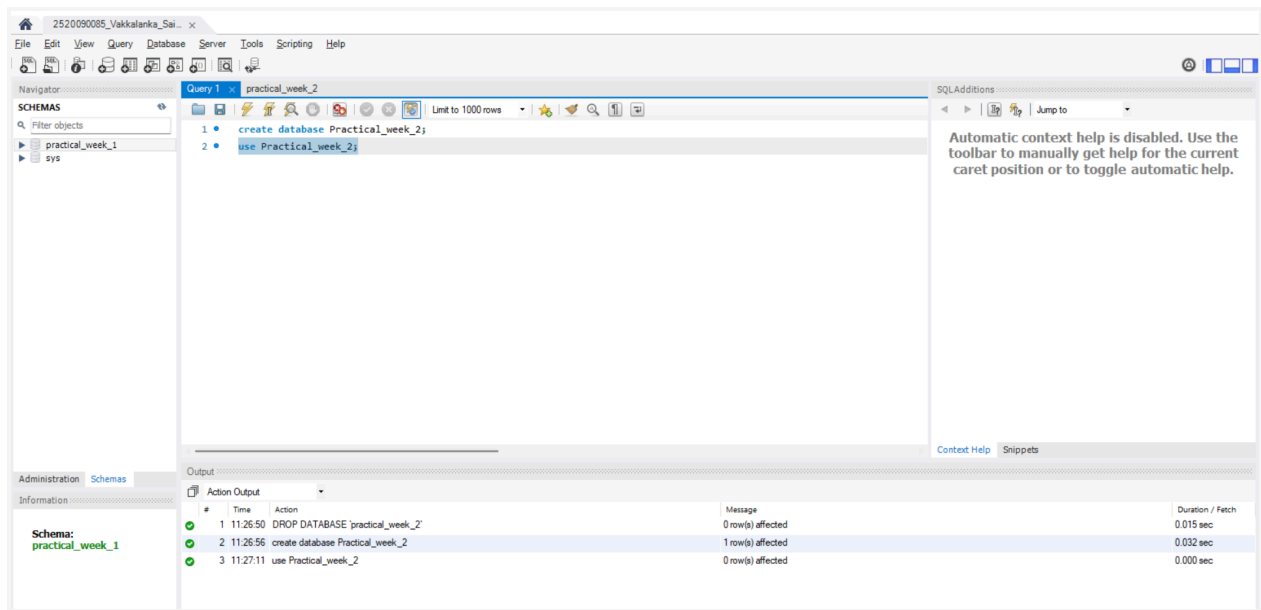


DBSE&DBD Practical Week 2

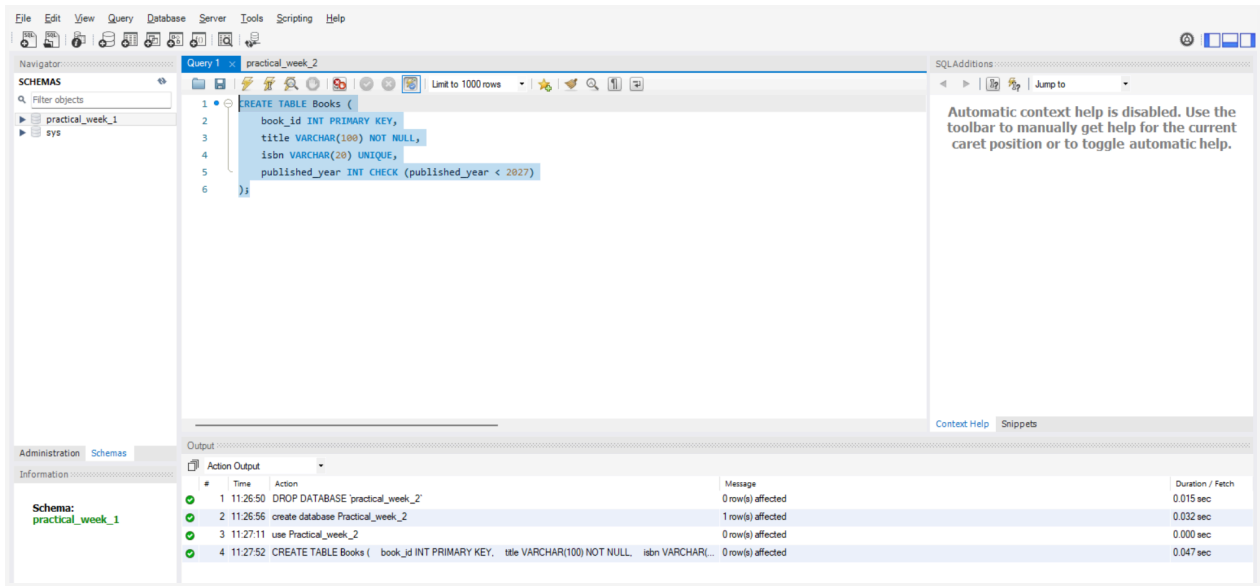
- Name: Vakkalanka Sai Bhaskara Sujit
- Roll No: 2520090085

SQL Queries and Expected Executions

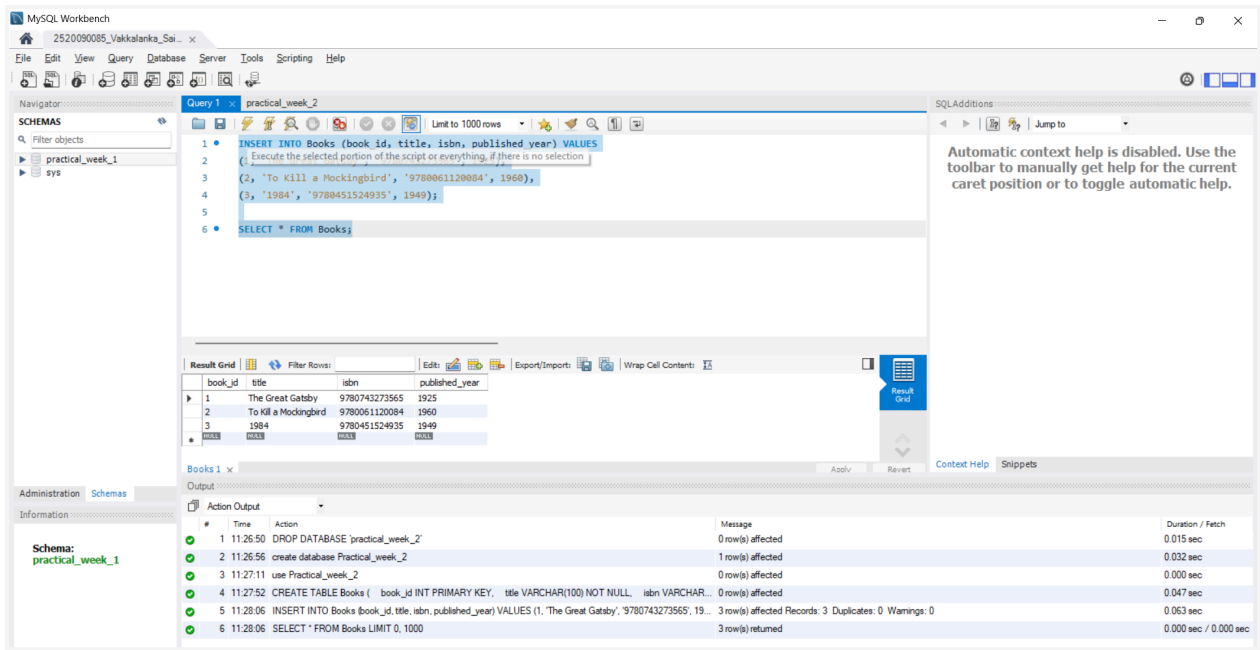
Step 1: Create Database



Step 2: Create Books Table (DDL + Constraints)



Step 3 & 4: Insert Books and Display



Step 5, 6 & 7: Members Table: Create, Insert, Display

MySQL Workbench

2520090085_Vakkalanka_Sai... x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

practical_week_1

sys

Query 1 practical_week_2

```

1 CREATE TABLE Members (
2   member_id INT PRIMARY KEY,
3   full_name VARCHAR(100),
4   email VARCHAR(100) UNIQUE
5 );
6
7 INSERT INTO Members (member_id, full_name, email) VALUES
8   (101, 'John Smith', 'john.smith@email.com'),
9   (102, 'Emma Wilson', 'emma.wilson@email.com'),
10  (103, 'Michael Brown', 'michael.brown@email.com');
11
12 SELECT * FROM Members;

```

Result Grid

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

Members 2

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	11:27:11	use Practical_week_2	0 row(s) affected	0.000 sec
4	11:27:52	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(100) NOT NULL, published_year INT)	0 row(s) affected	0.047 sec
5	11:28:06	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.063 sec
6	11:28:06	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
7	11:28:25	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	0 row(s) affected	0.078 sec
8	11:28:25	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma Wilson', 'emma.wilson@email.com'), (103, 'Michael Brown', 'michael.brown@email.com')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
9	11:28:25	SELECT * FROM Members LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec

Administration Schemas

Information

Schema: practical_week_1

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Step 8: Create Loans Table (FOREIGN KEYS)

MySQL Workbench

2520090085_Vakkalanka_Sai... x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

practical_week_1

sys

Query 1 practical_week_2

```

1 CREATE TABLE Loans (
2   loan_id INT PRIMARY KEY,
3   member_id INT,
4   book_id INT,
5   loan_date DATE,
6   FOREIGN KEY (member_id) REFERENCES Members(member_id),
7   FOREIGN KEY (book_id) REFERENCES Books(book_id)
8 );
9

```

Result Grid

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	11:27:52	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(100) NOT NULL, published_year INT)	0 row(s) affected	0.047 sec
5	11:28:06	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.063 sec
6	11:28:06	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
7	11:28:25	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	0 row(s) affected	0.078 sec
8	11:28:25	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma Wilson', 'emma.wilson@email.com'), (103, 'Michael Brown', 'michael.brown@email.com')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
9	11:28:25	SELECT * FROM Members LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
10	11:28:50	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE)	0 row(s) affected	0.047 sec

Administration Schemas

Information

Schema: practical_week_1

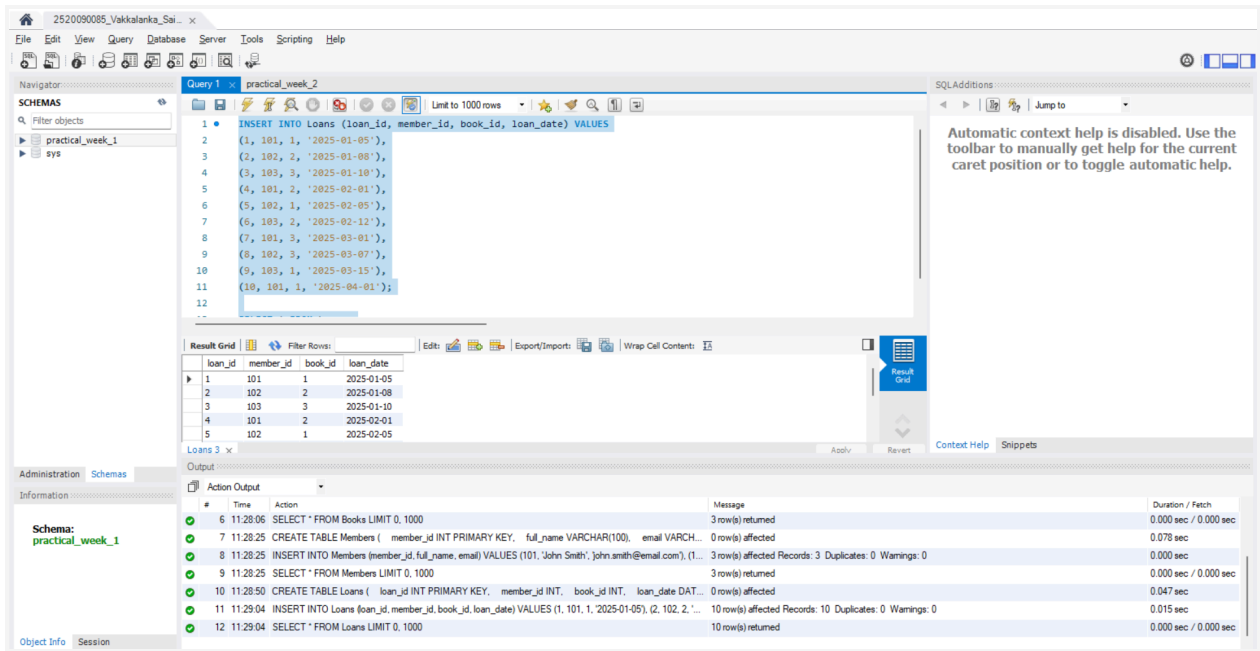
Object Info Session

SQLAdditions

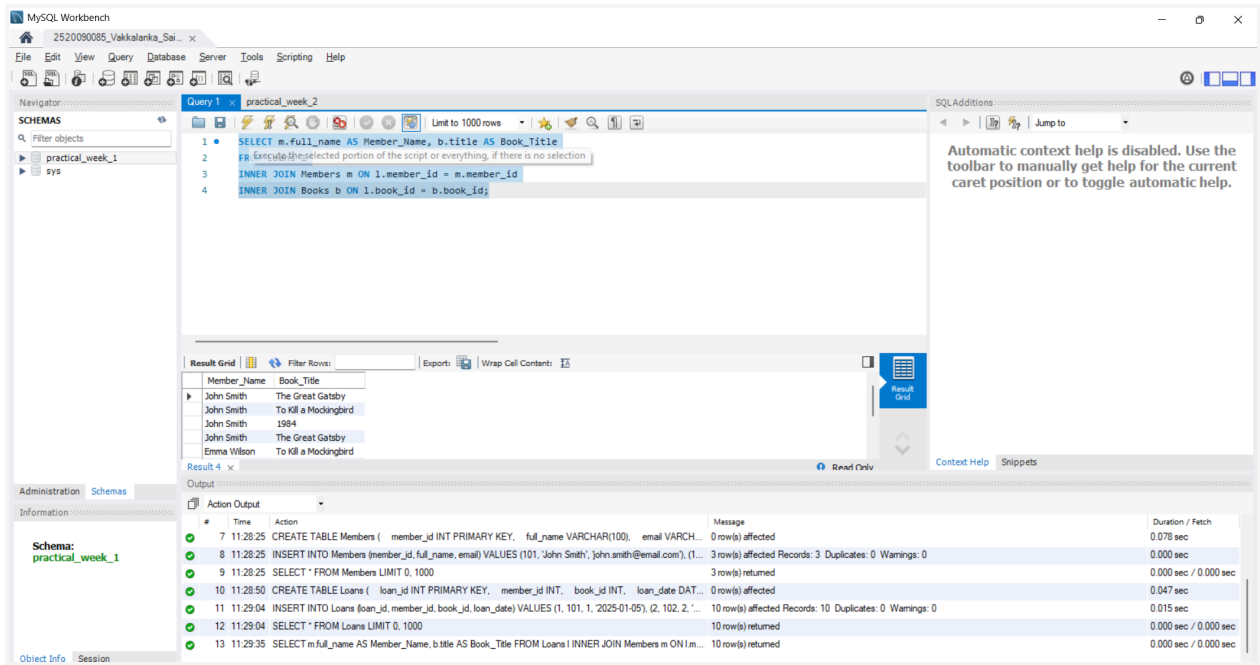
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

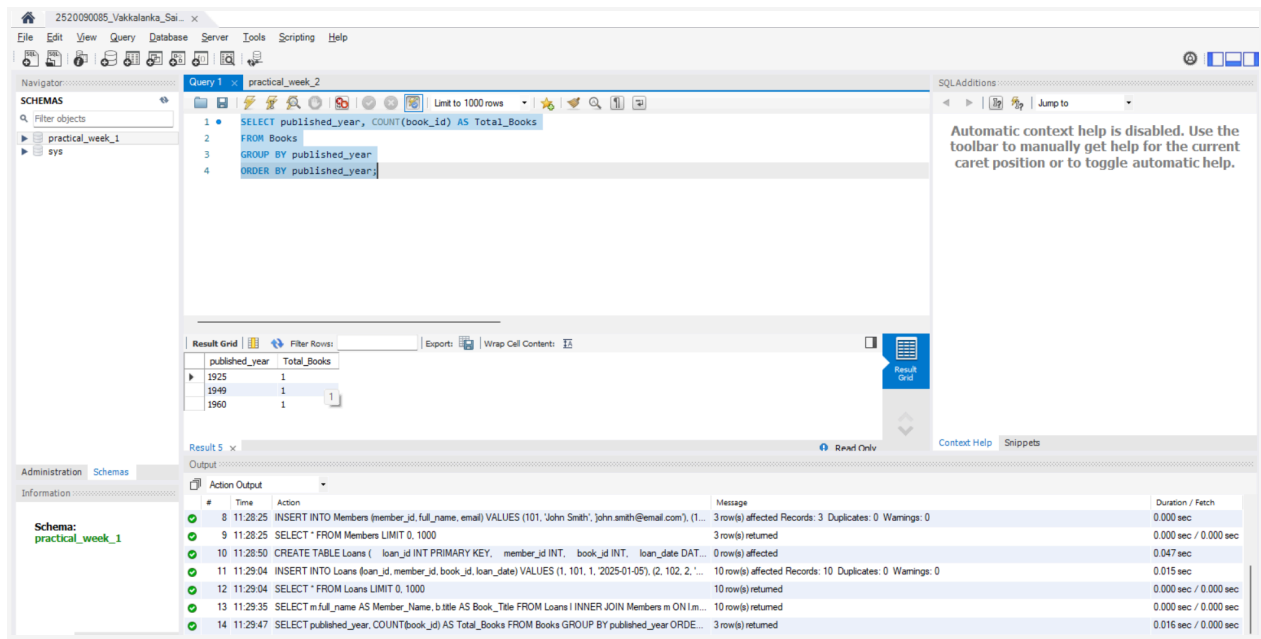
Step 9 & 10: Insert 10 Loans and Display



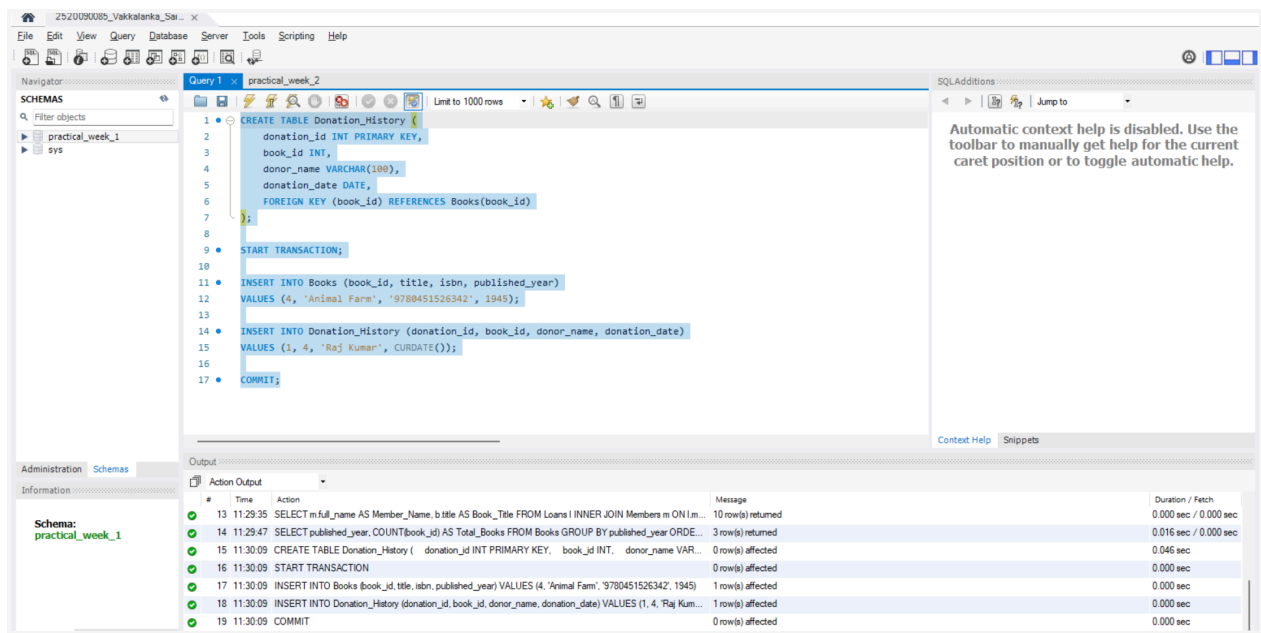
Step 11: JOIN Query



Step 12: GROUP BY Query



Step 13-16: Donation via TRANSACTION (Atomicity)



Step 17 & 18: Index on ISBN + Fast Search

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Navigator

SCHEMAS

Filter objects

practical_week_1

sys

Query 1: practical_week_2

1 CREATE INDEX idx_books_isbn ON Books(isbn);

2

3 SELECT * FROM Books WHERE isbn = '9780451524935';

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

book_id	title	isbn	published_year
3	1984	9780451524935	1949

Books 6

Apply Revert Context Help Snippets

Administration Schemas

Information

Schema: practical_week_1

Action Output

#	Time	Action	Message	Duration / Fetch
15	11:30:09	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VAR...	0 row(s) affected	0.046 sec
16	11:30:09	START TRANSACTION	0 row(s) affected	0.000 sec
17	11:30:09	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', '9780451526342', 1945)	1 row(s) affected	0.000 sec
18	11:30:09	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'Raj Kum...	1 row(s) affected	0.000 sec
19	11:30:09	COMMIT	0 row(s) affected	0.000 sec
20	11:30:33	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.109 sec
21	11:30:33	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec